

Ayan Nath

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EDUCATION

Massachusetts Institute of Technology
Ph.D. in Mathematics
Advisors: Bjorn Poonen, Keerthi Madapusi

Cambridge, MA
September 2024 – present

Chennai Mathematical Institute
Bachelor of Science (Honours) in Mathematics and Computer Science

Chennai, India
September 2021 – April 2024

RESEARCH

- **Perfectoid Shimura Varieties of Exceptional Type** (draft) 2026
We show that all minimally compactified Shimura varieties at infinite level, including those of exceptional type, are perfectoid spaces for all sufficiently large primes.
- **Compactification of Reductive Group Schemes** [[arXiv](#)] 2026
We prove that any isotrivial reductive group scheme over an arbitrary base admits a smooth projective equivariant compactification, verifying a conjecture of Česnavičius. In the adjoint case, we recover fiberwise the wonderful compactification. Finally, we give an example of a non-isotrivial torus admitting no equivariant compactification.
- **The Moments of Split Greatest Common Divisors** (with Abhishek Jha and Emanuele Tron) 2024
to appear in Acta Arithmetica [[arXiv](#)]
- **On the Least Common Multiple of Polynomial Sequences at Prime Arguments** (with Abhishek Jha) 2022
International Journal of Number Theory, 18(06), [[doi](#)]
- **On Quotients of Values of Euler's Function on Factorials** (with Abhishek Jha) 2021
Bulletin of the Australian Mathematical Society, 105(3), [[doi](#)]
- **On the divisibility $a! + b! \mid (a + b)!$** 2022
The American Mathematical Monthly, 129(3), [[doi](#)]

SERVICE

- **Research Science Institute (RSI), MIT** Summer 2026
Mentor for two high school students pursuing original research projects.
- **Directed Reading Program, MIT** January 2026
Mentor for three undergraduate students on complex elliptic curves and their moduli.
- **Directed Reading Program, MIT** January 2025
Mentor for two undergraduate students on the Kronecker-Weber theorem and Artin reciprocity.
- **Teaching Assistant, Chennai Mathematical Institute**
 - **Calculus 1** (Multivariable differential calculus) January 2024 – April 2024
 - **Calculus 2** (Multivariable integral calculus) August 2023 – December 2023

- **Analysis 2** (Topology, function spaces, Fourier analysis)
- **Discrete Mathematics**

August 2023 – December 2023

January 2023 – April 2023

AWARDS

- 2024 **Akamai Presidential Graduate Fellowship**, MIT.
CMI Medal of Excellence, top of class.
- 2022 **SRIRAM Scholarship**, tuition fee waiver and monthly stipend for undergraduate studies.
Spirit of Ramanujan
Indian National Mathematical Olympiad Awardee (2019, 2020, 2021), Homi Bhabha Centre For Science Education